Hima Shanthi Rayapati

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Education

Arizona State University, Data Science, Analytics and Engineering (Human-Centered Applications) 08/2026 – 12/2027

- Fulton Schools of Engineering Graduate Scholarship | ASU AI Scholars Program, 2026–27

Arizona State University, BS in Psychology and Mathematics (Statistics) 01/2023 – 05/2026

- **Minor:** Philosophy | **GPA:** 4.0/4.0

Research Experience

Research Assistant, *CARE Lab (Prof. Qiaoning Zhang)*, ASU 08/2026 – Present

- Attend lab meetings and discuss ongoing research on human-AI interaction.
- Building a multimodal AI tutor where students choose how they learn (visual, text, or speech), to use in a study comparing it with a standard AI tutoring chatbot (in progress).

UX Researcher (UXR Guild), *NEXT Lab*, ASU 08/2026 – 10/2026

- Tested games, digital twins, and interactive apps with users to find usability problems and collect feedback.

Research Assistant, *Thinking Across Languages and Contexts Lab*, ASU 08/2025 – 10/2026

- Assisted a graduate student with testing experiments built in jsPsych and JavaScript.
- Helped check whether data were recorded correctly and organized for analysis.
- Performed exploratory data analysis and assisted in interpreting experimental results under supervision.

Research Assistant, *Perception, Action, Language & Meaning Lab*, ASU 08/2025 – 05/2026

- Ran two-person collaborative task sessions (e.g., Taboo), observing and recording coordination, communication, and performance data using VR, motion-capture, and eye-tracking.
- Worked with the lab's time-series behavioral data, learning how to clean, organize, and explore it.
- Read papers and explored possible methods (e.g., multilevel vector autoregression) to understand which approaches might fit the lab's data.
- Contributed to discussions by sharing preliminary notes and observations from trial analyses.

Volunteer Research Assistant, *Longitudinal & Behavioral Data Science Lab*, ASU 08/2024 – 12/2024

- Participated in bi-weekly lab meetings, contributing to discussions and staying updated on research progress.

Research Assistant, *Evolution, Ecology, and Social Behavior Lab*, ASU 08/2023 – 11/2023

- Assisted with survey design, pilot testing, data coding, and literature reviews in social cognition research.
- Participated in lab meetings and contributed to discussions on study design and interpretation.

Projects

Carryover: AI-Supported Daily Living App for Autistic Adults ***GitHub***

Tools: Python, React, FastAPI, pytest · **Methods:** User research, Task analysis, Literature review, Rule-based prompt fading

- Led user research with partner First Place AZ (a site visit and 2 staff meetings), and turned the findings into design changes, including always-visible safety cautions and adaptive help per step.
- Wrote a literature review on prompt fading and executive function, and wrote the fading rules (when help fades or comes back, per step and per person) that the app follows.
- Built a Python module that detects where users get stuck (4 patterns) from interaction logs, requiring patterns to repeat before acting, to avoid false alarms.
- Co-built a working prototype with simulated data (React, FastAPI, an LLM with code-based safety rules, and automated tests) on a 5-person team, and presented it at the Scottsdale Launchpad.

Trust in Advice: AI vs. Human Sources (Data Analysis & Written Report) ***Trust in AI.com/2025***

Tools: SAS (PROC GLMSELECT, PROC GLM) · **Methods:** General linear model, Stepwise selection (AIC), Residual diagnostics

- Analyzed 30,000+ human-AI interactions to study effects of advice source.
- Modeled decision changes before and after advice exposure.
- Found advice correctness as primary driver, with modest AI-related effects.

Analyzing the Impact of AI on Stack Overflow Participation (NLP & Data Analysis) *GitHub*

Tools: Python (pandas, NumPy, statsmodels, SciPy, Matplotlib, Seaborn), SQL, Jupyter · **Methods:** Time-series analysis, NLP

- Analyzed large-scale Stack Overflow data using Python to study changes in user behavior after AI tool releases.
- Built time-series datasets and computed engagement metrics (answer rates, response time, retention) using Python.
- Performed statistical comparisons and visualized platform-level trends.
- Explored tag-based trends and basic NLP features to examine topic shifts.

Cultural Differences in Trust: AI vs. Human Expert Decision-Making *Research Proposal / 2025*

Methods: Mixed-effects modeling, Experimental design, Pre-registration

- Proposed cross-cultural study (Mexico vs. U.S.) on AI vs. human decision-making under conflict.
- Outlined experimental design focusing on competence and legitimacy.
- Formulated mixed-effects modeling approach for analysis.
- Synthesized literature across cognition, decision-making, and human-AI interaction.

Detecting and Correcting Social Bias in AI Question Answering (CSE 575, in progress)

Methods: Text classification, Bias evaluation, Data preprocessing

- Team project building a pipeline that detects biased answers from LLMs on the BBQ bias benchmark and rewrites them, while keeping accuracy on questions the context can answer.
- Responsible for the data side: auditing and preprocessing BBQ, a loader that converts it into a general (text, label) format, and template-grouped train/validation/test splits so near-duplicate questions don't leak into the test set.

Stochastic Dual-Timescale Model of Learning (ODE & SDE Modeling) *Final Project / 2025*

Tools: MATLAB, R Shiny · **Methods:** ODE/SDE modeling, Stability analysis, Simulation

- Developed differential equation model of fast and slow learning processes.
- Extended to stochastic framework to examine effects of noise.
- Verified theoretical properties via simulation.
- Linked model behavior to memory retention and learning variability.

Probabilistic Reward Learning Task using PsychoPy *Reward Learning.com/2025*

Tools: PsychoPy, Python (pandas, NumPy) · **Methods:** Experimental design, Learning under uncertainty

- Designed probabilistic reward-learning task with informative vs. non-informative features.
- Examined learning under uncertainty and perceptual noise.
- Collected and analyzed participant data using Python.

Predictive Analysis of Team Performance in Baseball using SAS *Baseball Analysis.com/2024*

Tools: SAS · **Methods:** Multiple regression, EDA, Correlation analysis

- Developed a statistical model to predict team wins using historical baseball data.
- Conducted EDA and correlation analysis on Minnesota Twins batting records (1998-2024, excluding 2020).
- Built multiple regression models to identify key predictors of team wins, selecting runs and batting averages for a parsimonious model.
- Addressed multicollinearity and validated model assumptions to improve predictive accuracy.

Predicting Daily Stock Prices of ADP using SAS *ADP Stock.com/2024*

Tools: SAS · **Methods:** Multiple regression, Stepwise selection, EDA

- Created a predictive model for ADP stock prices based on financial indicators.
- Applied EDA, correlation analysis, and multiple regression (stepwise, backward, forward selection) to determine significant predictors.
- Developed a quadratic regression model with strong explanatory power while addressing multicollinearity.
- Examined sector-specific dynamics and market trends influencing ADP's stock behavior.

Cyber Attack Detection in Water System Sensor Data (CSE 575, in progress)

Methods: Anomaly detection, Time series, Feature engineering, Ridge regression

- Group project using the BATADAL water distribution system dataset to detect cyber attacks from sensor readings.
- Did data cleaning and auditing, feature engineering, a Ridge regression baseline, and handling of partly labeled data.

Research Proposal: Mirror Neurons, Social Synchrony, and Collaborative Learning ***Proposal.com/2025***

- Proposed a research framework synthesizing human and animal literature on mirror neuron systems and social synchrony.
- Formulated hypotheses linking coordination mechanisms to collaborative learning outcomes.

Additional projects and full write-ups available at my portfolio.

Work Experience

Graduate Course Assistant (Statistical Methods), ASU – Tempe, AZ 10/2026 – Present

- One of two course assistants for PSY 330 – Statistical Methods.
- Grade assignments and exams with written feedback, following the instructor's grading guidelines.
- Answer student emails about course content and SPSS.

Research Aide, ASU – Tempe, AZ 12/2024 – Present

- Review, categorize, and structure user-generated data to ensure quality and consistency.
- Assign relevant tags to comments for improved organization and searchability.
- Check whether AI-generated tags follow the evaluation guidelines and are accurate.
- Perform data cleaning tasks, identifying and correcting inconsistencies to maintain data accuracy.
- Work with databases to structure data and ensure accessibility for future use.

Lead Survey Assistant, ASU – Tempe, AZ 05/2023 – 08/2026

- Conducted phone-based survey interviews with currently employed and recently graduated ASU students, ensuring accurate and complete data collection by clarifying responses and addressing any ambiguities.
- Assisted in the hiring process, including interviewing and evaluating candidates.
- Performed data cleaning and ensured the integrity of datasets for analysis.
- Trained new student workers in survey procedures and oversaw testing and calibration of survey tools.

Intern, Public Safety Crisis Solution – Phoenix, AZ 08/2024 – 11/2024

- Performed quantitative analysis to support research, program development, and accreditation documentation.
- Contributed research-based content on clinical procedures addressing first responders' mental health initiatives.

Advising Office Aide, ASU – Tempe, AZ 08/2023 – 05/2024

- Supported advising operations through scheduling, communication, and administrative workflows.
- Managed data entry, marketing/outreach materials, and front-desk support for students and staff.

Skills

- **Programming & Statistical Tools:** R, SAS, MATLAB, Python (pandas, NumPy, statsmodels, scikit-learn, Matplotlib), JMP, JASP, SQL (Microsoft Access), Java, C, Bash, REST APIs
- **Machine Learning, NLP & AI Evaluation:** scikit-learn, Classification (logistic regression, random forest, SVM), Ridge regression, Clustering, Dimensionality reduction (PCA), Neural networks (fundamentals), Feature engineering, Hyperparameter tuning, Cross-validation, Model evaluation (precision/recall/F1, ROC-AUC), Text classification, Basic NLP, LLM output evaluation, Human-AI agreement analysis, Bias and fairness evaluation, Anomaly detection, Time-series analysis
- **Data Analysis & Visualization:** Tableau, Excel, Google Sheets, Alteryx
- **Experimental & Research Tools:** jsPsych, PsychoPy, JavaScript, HTML, CSS, Qualtrics, SurveyMonkey
- **Software & Development:** React, FastAPI, pytest, Git/GitHub
- **Design & Prototyping:** Figma, Figma Motion
- **Collaboration & Productivity:** Slack, Teams, Zoom, Word, Outlook, Adobe PowerPoint, OneDrive, Dropbox